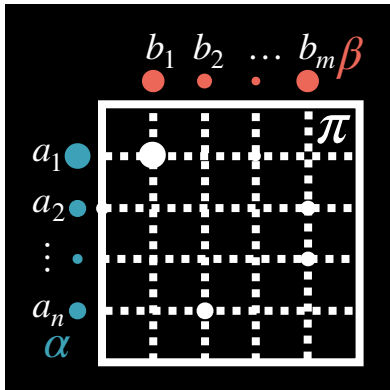
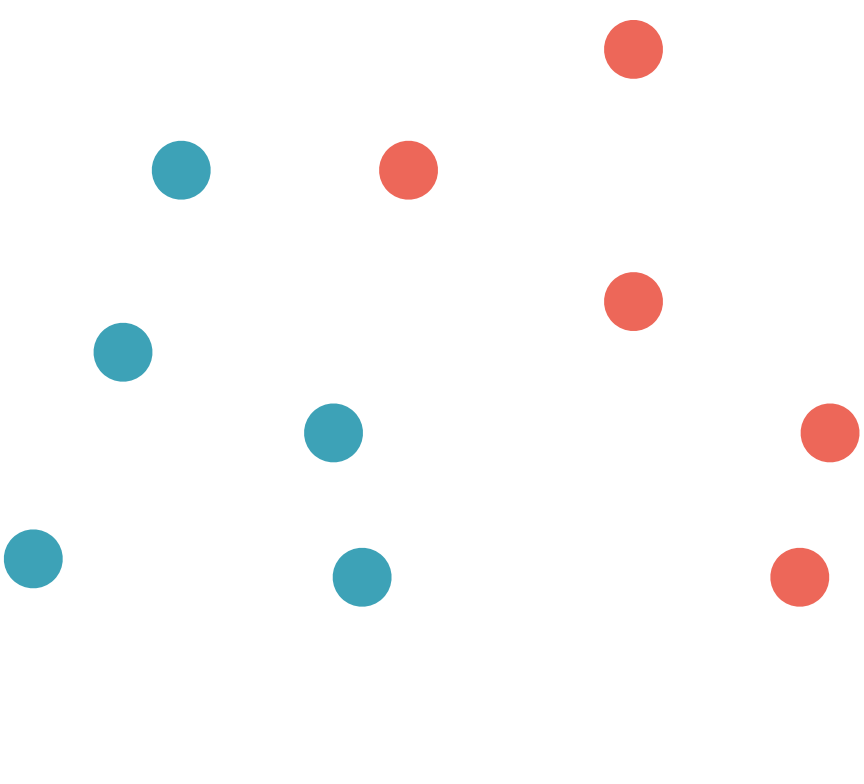
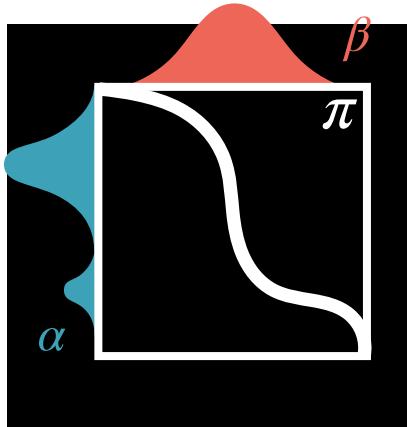
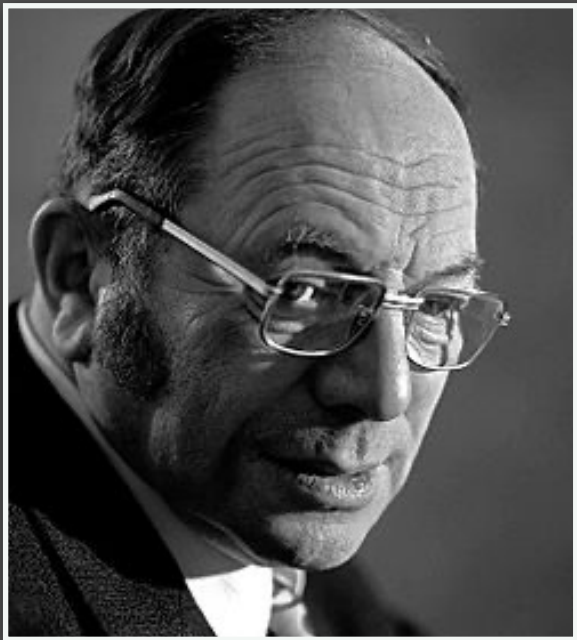






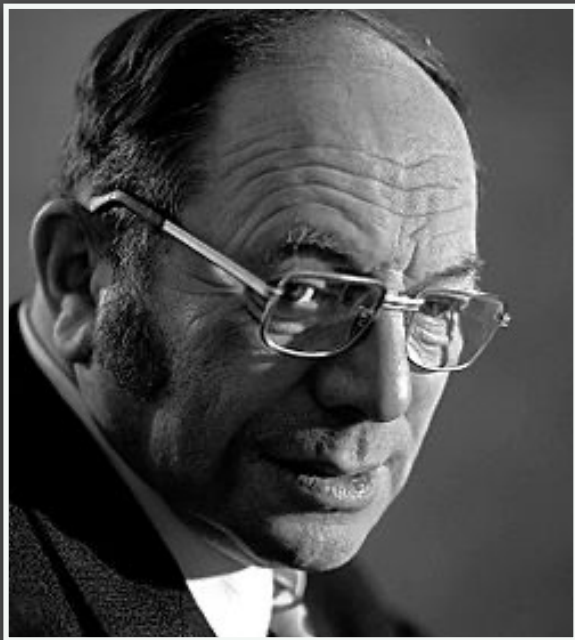




Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m \mathbf{C}_{i,j} \pi_{i,j}$$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$

In both cases: can interpret as “average” work to move mass

#3D251F
cocap

#3DA2B7
teal

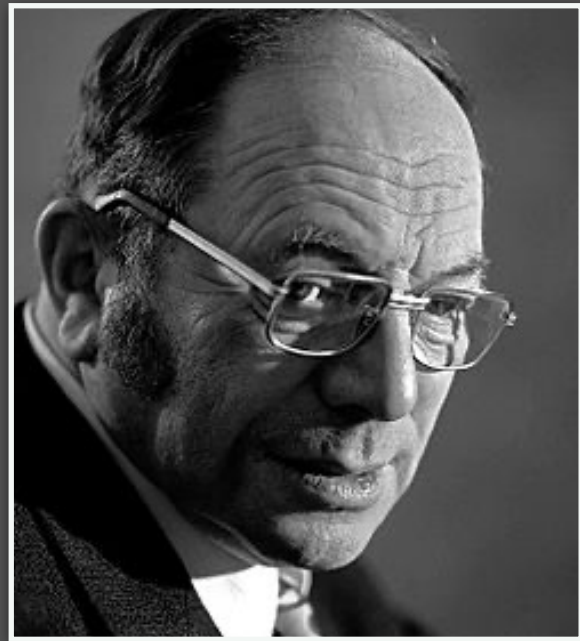
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salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

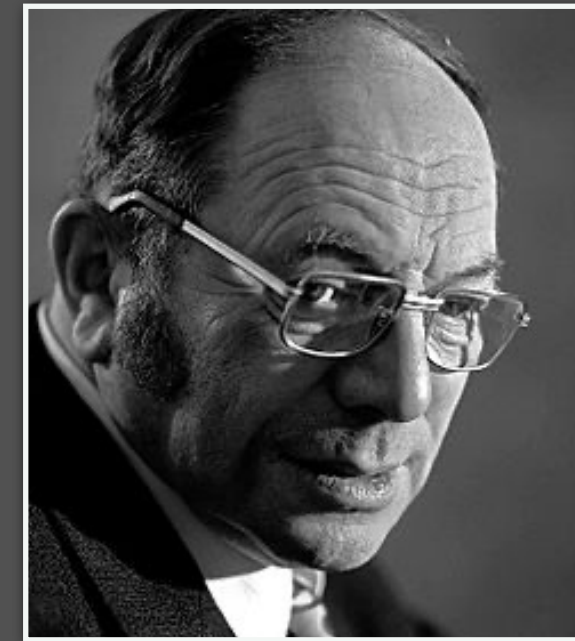
Kantorovich Formulation: General Case



Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m C_{i,j} \pi_{i,j}$$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

Measures

$$\alpha = \sum_i a_i \delta_{x^{(i)}}, \beta = \sum_j b_j \delta_{y^{(j)}}; \quad \mathbf{a} \in \Delta_n, \mathbf{b} \in \Delta_m$$

$$\alpha \in \mathcal{P}(\mathcal{X}), \beta \in \mathcal{P}(\mathcal{Y})$$

Cost

$$\mathbf{C} \in \mathbb{R}^{n \times m}, \quad C_{ij} = c(x^{(i)}, y^{(j)})$$

$$c(\cdot, \cdot) : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$$

Coupling

$$\pi = \begin{array}{c} \begin{array}{c} b_1 \quad b_2 \quad \dots \quad b_m \end{array} \beta \\ \begin{array}{c} a_1 \quad a_2 \quad \dots \quad a_n \end{array} \alpha \end{array} \begin{array}{c} \pi \\ \begin{array}{c} \cdot \\ \cdot \\ \cdot \\ \cdot \end{array} \end{array} \in \mathbb{R}^{n \times m}$$

$$\pi = \begin{array}{c} \beta \\ \alpha \end{array} \begin{array}{c} \pi \\ \cdot \end{array} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$$

Problem

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \Gamma, \mathbf{C} \rangle = \sum_{ij} \pi_{ij} c(x^{(i)}, y^{(j)})$$

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y)$$

Constraints

$$\Pi(\mathbf{a}, \mathbf{b}) = \{ \pi \mid \pi^\top \mathbf{1} = \mathbf{a}, \pi \mathbf{1} = \mathbf{b} \}$$

$$\Pi(\alpha, \beta) = \{ \pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \mid P_{\mathcal{X}\#} \pi = \alpha, P_{\mathcal{Y}\#} \pi = \beta \}$$

In both cases: can interpret as “average” work to move mass

Three types of couplings

